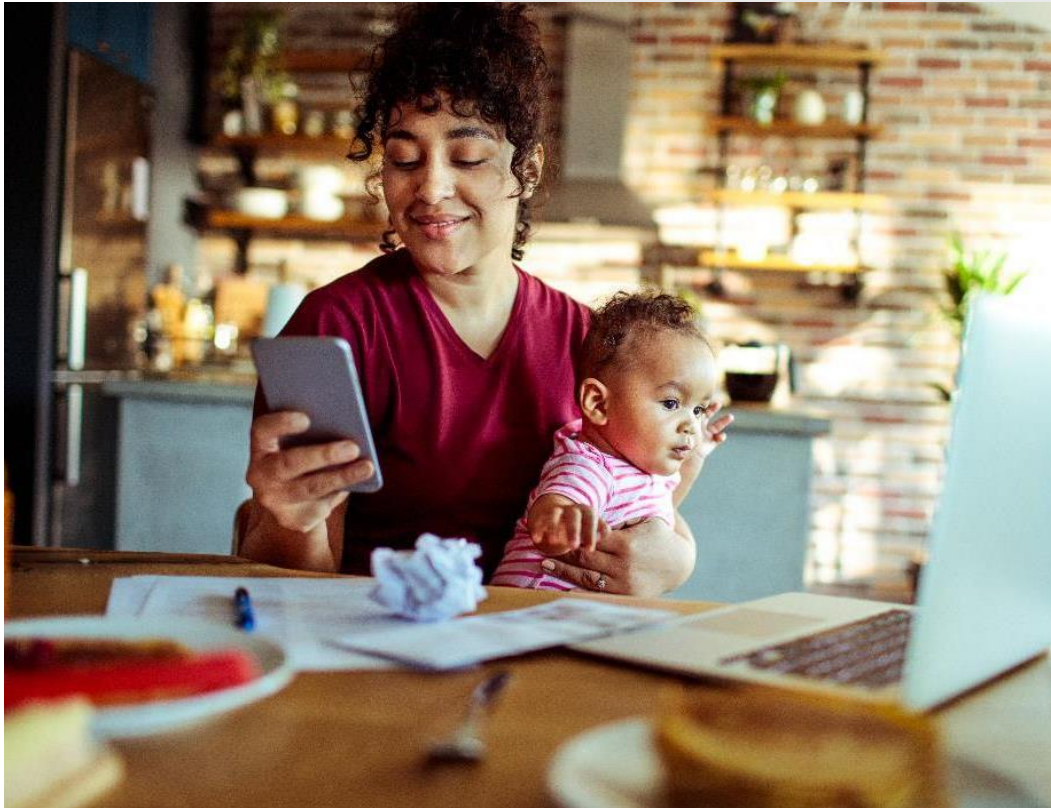




i90 – Aggregation Framework

([Rally Epic - E16777](#) & [slack channel](#) - [#i90-digital-community-of-practice-team-rx-cop](#))

July 12, 2023



Agenda

i90 Aggregation Framework – Goals & Architectural Runway (5 minutes)

Demos -From Production – (5 Minutes)

1) Demo - Schema Modelling

Schema Design – Demo (Hackolade Tool)

In Production View– Apollo Voyager

2) Demo-Experience APIs Portal

3) Demo -Observability – VALET and DORA Metrics

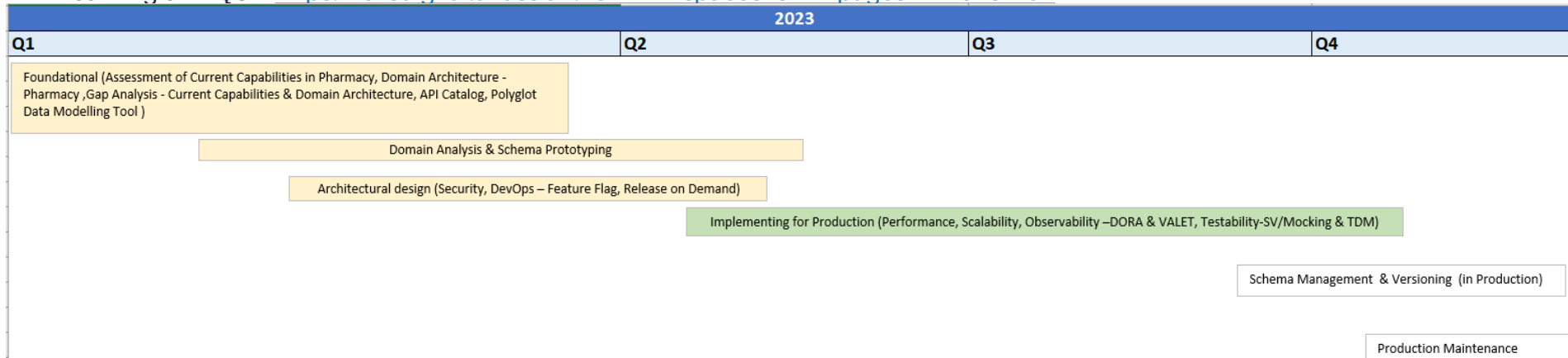
i90 Aggregation Framework – Goals & Architectural Runway

Goals – Simplify

- Simplify Software Design & Development using Domain Driven Design (DDD) approach .
- Organizing the software Design & Development around Experiences and Capabilities for Business Domains rather than building adhoc APIs needed for business features. Decouple front end and back end.
- Identify Orphan and/or duplicate Business capabilities
- Align to Domain Architecture - <https://cvsdigital.atlassian.net/wiki/spaces/ARCH/pages/2545124676/Domain+Architecture>

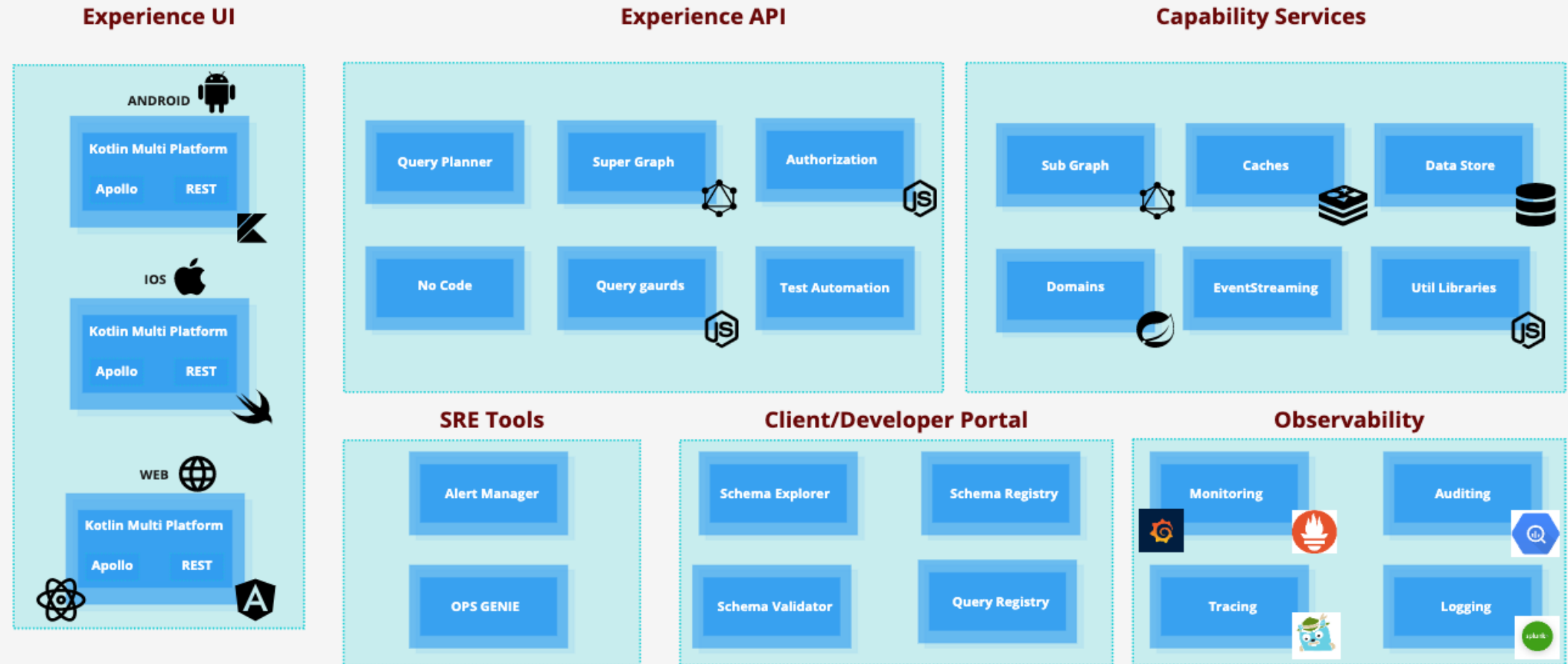
Architectural Runway- <https://cvsdigital.atlassian.net/wiki/spaces/CDH/pages/2701230954/Epic+16777+Aggregation+API+Framework+Backlog+Architecture+Runway>

- Learning & FAQs - <https://cvsdigital.atlassian.net/wiki/spaces/CDH/pages/2776434192>

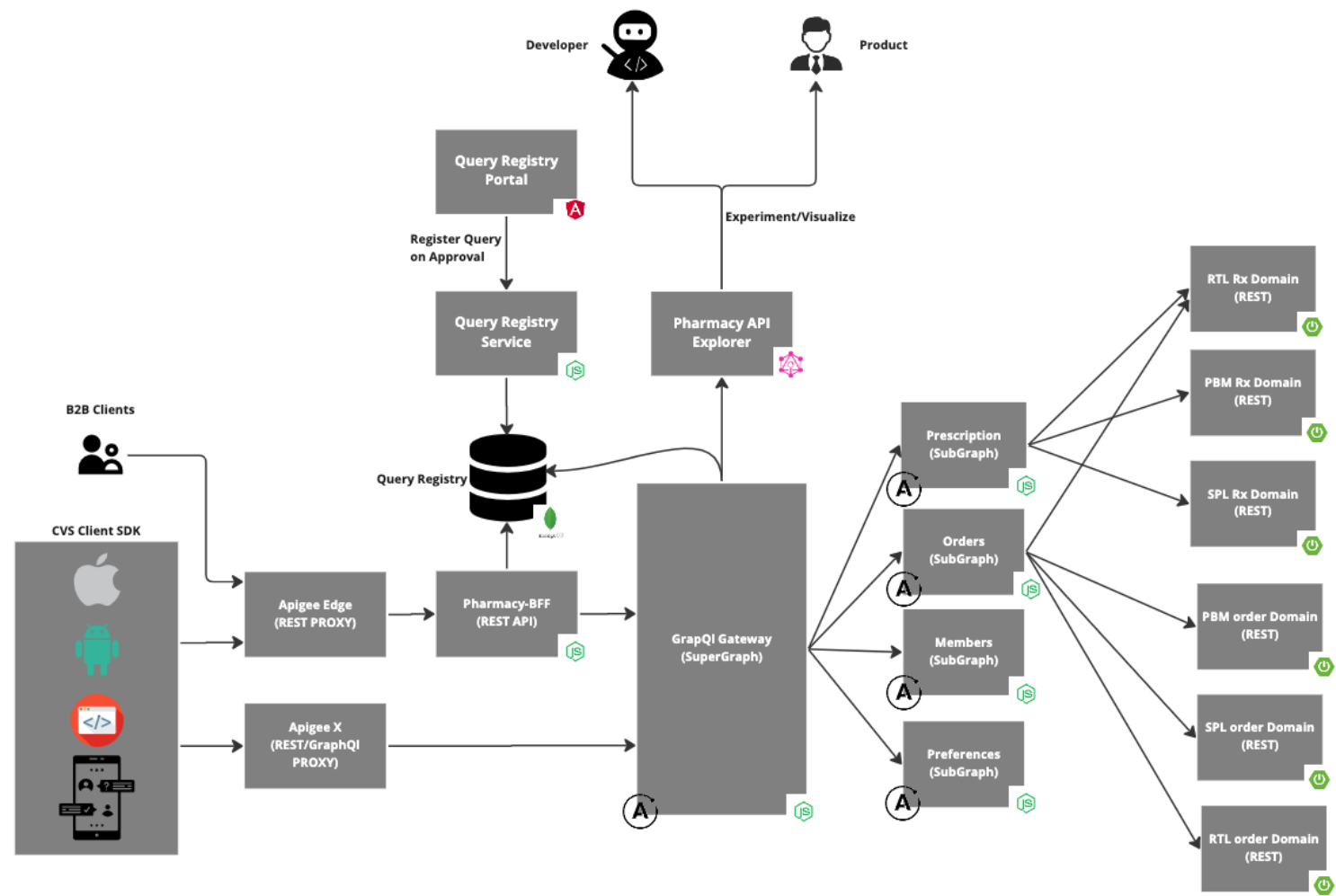


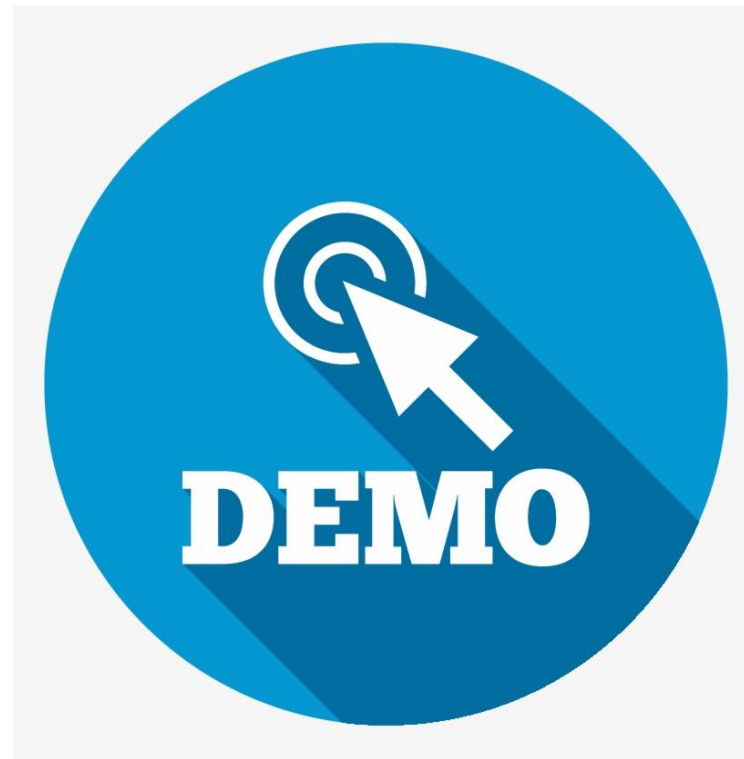
Legend of Activities : Green - In progress, Yellow- Completed(Draft Ready), None- Not started yet

Aggregation Framework – Approach



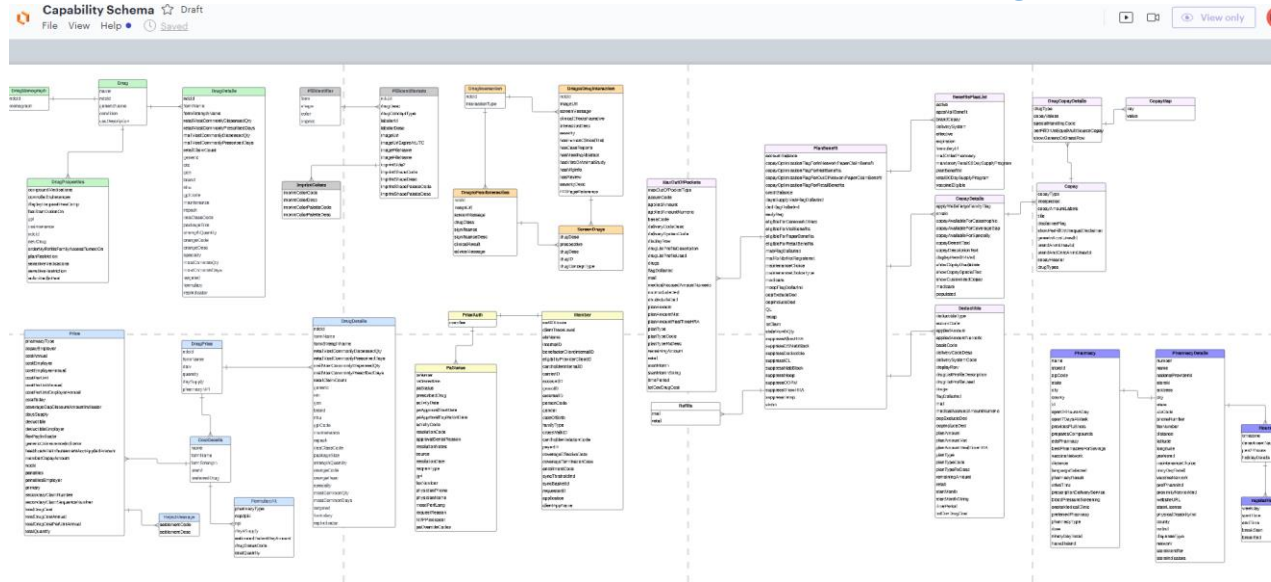
i90- Conceptual View





Demo – Schema Modeling (Hackolade Tool)

<https://hackolade.com/domain-driven-data-modeling.html>



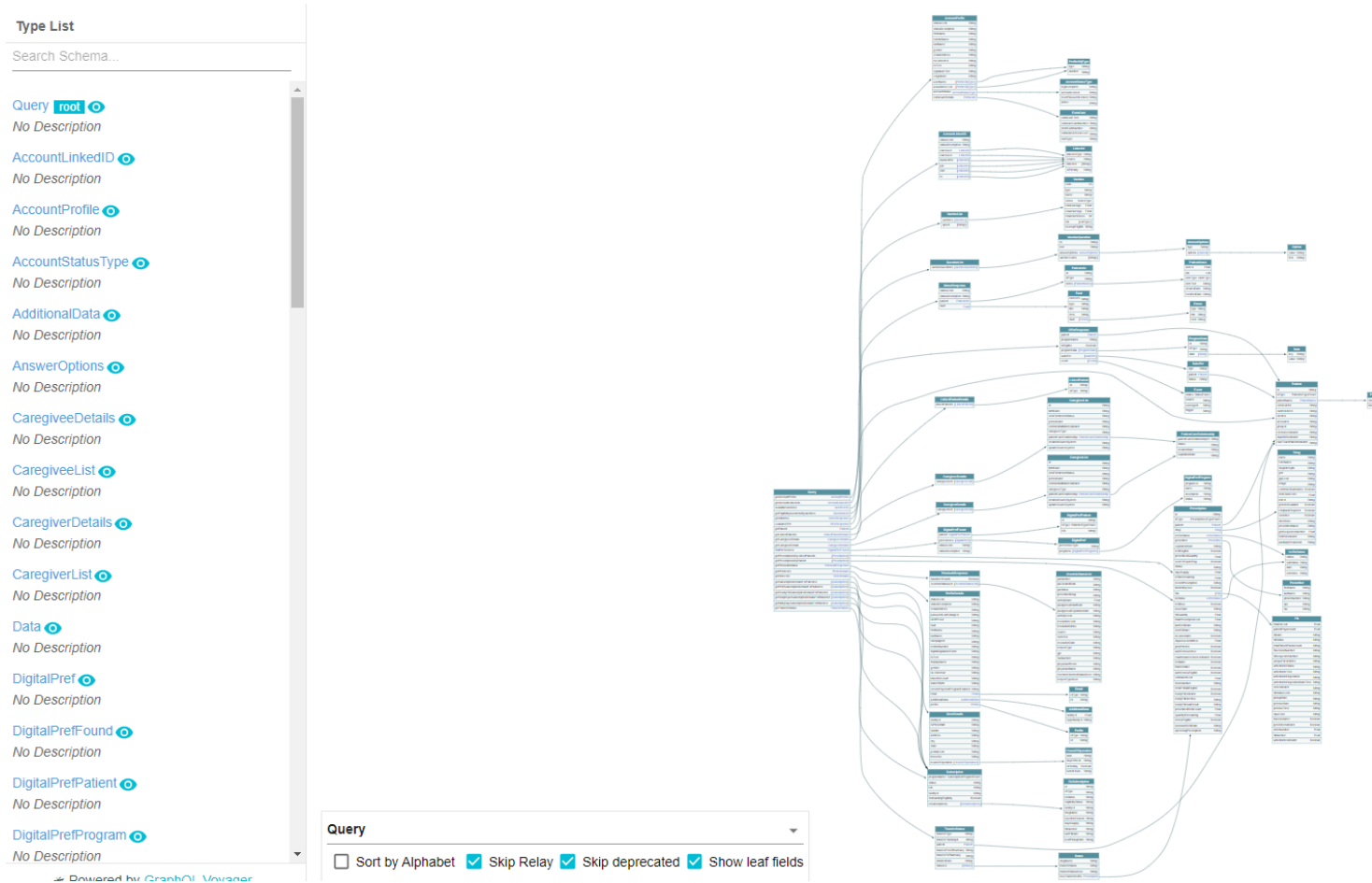
Sample (Pharmacy domain): https://lucid.app/lucidchart/18ef50d7-780f-4b70-9f98-fbb95905d180/edit?invitationId=inv_4617b87b-5e00-4bde-a1a7-b53f18a6b216&page=4~v_IdiXhUFq#

Pharmacy schemas and Domains and subdomains - Tracking

<https://cvsdigital.atlassian.net/wiki/spaces/CDH/pages/2730623669/Tracking+GraphQL+schema+design>

Demo – In Production Domain Schema View– Apollo Voyager

<https://digital-retail-rx-qa.internal.cvshealth.com/voyager/>



Demo - Experience APIs Portal (in Production)

<https://www.cvs.com/experience/intake/api/view-exp>



Add New API

Experience Name	App Name	Line Of Business	API Type	Status	View Details	VALET Metrics	Alerts
OnboardingExperienceApi	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts
chatbot_refill_initial_info	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts
Retail_Login_Prefetch	Retail	Front Store	Rest	LIVE	View Detail		Alerts
SubscriptionStatusVone	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts
SubscriptionEnrollMentExpVOne	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts
AvailableVaccines	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts
PHREnrollmentMutationVOne	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts
PHRSubscriptionStatusVone	Pharmacy	Retail Pharmacy	Rest	LIVE	View Detail		Alerts

1 to 10 of 14<< < Page 1 of 2 > >>

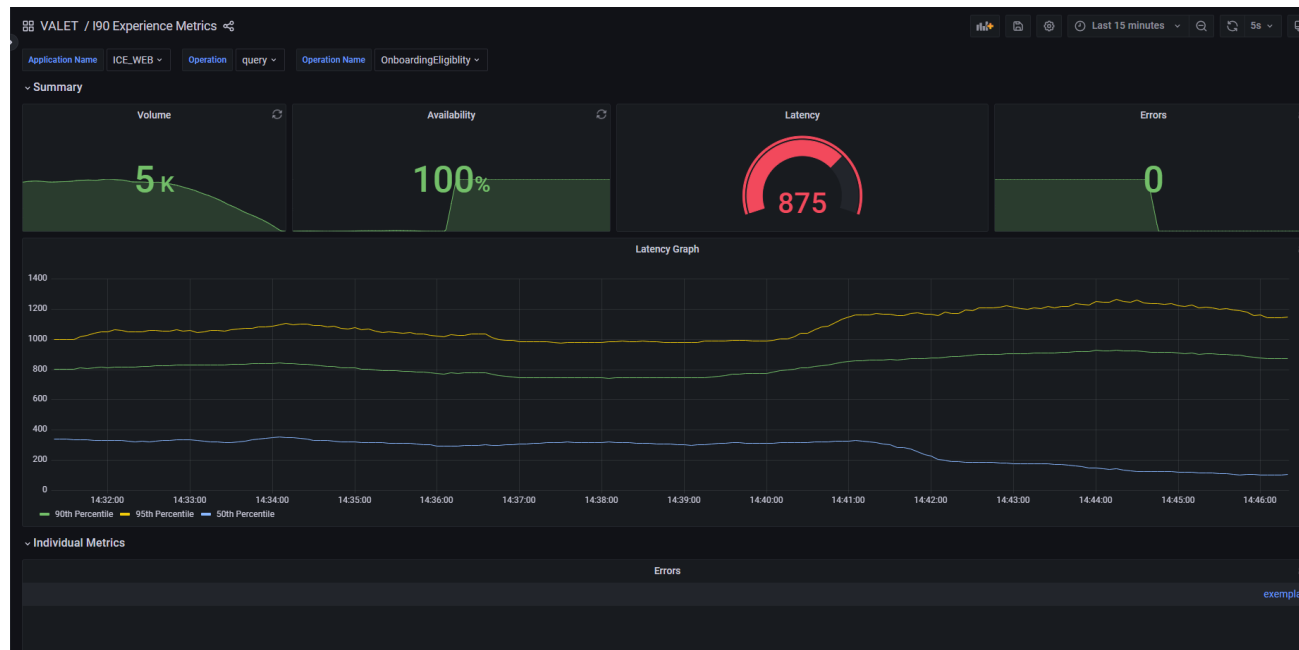
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Observability – VALET and DORA Metrics (in Production)

VALET - https://grafana.digital-ice-prod.internal.cvshealth.com/d/graphql/i90-experience-metrics?orgId=1&refresh=5s&var-app=ICE_WEB&var-operation=query&from=now-15m&to=now&var-operationName=OnboardingEligibility

DORA - <https://digital-fourkeys.cvshealth.com/d/opbZSuSVz1/dora-metrics-service-dashboard?orgId=1>



Latest news in on slack - APIs are constantly being deployed in Production ; New features are constantly being added to Framework

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I90-Experience Valet M... I90 Experience - Registr... I90 Experience Best Pra... +

Today

Devi Prakash Patti 1:37 PM

Hi Team, just fyi, following Domain and Capability API's have been deployed to production.

image.png

S.No	Feature	API	Deployment Status
1	get Member Info (patient) -PBM	Domain API	Deployed in prod
2	get Member Info (patient) -PBM	SubCapability API	Deployed in prod
3	get Member Info (patient) -SPL	Domain API	Deployed in prod
4	get Member Info (patient) -SPL	SubCapability API	Deployed in prod
5	get Plan details - PBM CSK	Domain API	Deployed in prod
6	get Plan details - PBM CSK	SubCapability API	Deployed in prod
7	get Member (Profile)-PBM	Domain API	Deployed in prod
8	get Member (Profile)-SPL	Domain API	Deployed in prod
9	Family Access	Domain API	Deployed in prod
10	Family Access	SubCapability API	Deployed in prod
11	Security options for Family CUID	Domain API	Deployed in prod

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1 reply 22 hours ago Yesterday

Rubesh Kumar Maniprakasam 4:52 PM

PBM-Prescription domain wrapper service is deployed in PROD [without Prescription History (as of now history is loaded from claims and order is not included)]

3 5 replies Last reply 18 hours ago

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Friday, June 2nd

i90-digital-community-of-practice-team-rx-cop Experience API, BFF Adoption 243

Yesterday

Gaurav Tyagi 2:52 PM

Capability APIs exception handling - Hi All, please find the draft for exception handling guidance for our capability APIs. Please review and provide feedback. You may reach out to @MrityunjayKumar, Fnu or @Tyagi, Gaurav for any questions. <https://cvsdigital.atlassian.net/wiki/spaces/CDH/pages/2868315205/I90+Graphql+ErrorHandling>

I90 Graphql ErrorHandling in CVS Digital Home

Last updated a day ago by Fnu Mrityunjay Kumar

Like page Watch More actions...

Added by Confluence Cloud

2 1 1 reply 24 hours ago

Dharmendra Gudimetla 4:16 PM

DYK Today, we have implemented a prefetch using the deep learning algorithm within the I90 platform. With this, pre Onboarding experience response time has been reduced from staggering 4 sec to under 500 ms as a result of all I90 framework optimization. This is a great story to celebrate as this leads for better customer experience.

Thanks to all contributors @Sanwal, Gaurav @Palani, Muthukumar @AnandaPadmanaban, Murali @Katare, Srikanth (Contractor) @Chilamkuri, Manohar (Contractor) @Achala, Lakshminarayana (Contractor) @Lal, VinitSathish @Ashokan, Anand @Shah, Atur

Reference of implementation <https://cvsdigital.atlassian.net/wiki/spaces/CDH/pages/2802386190/I90+Prefetch+Design+Pattern+-+DL+Model>

I90 Prefetch Design Pattern - DL Model in CVS Digital Home

Last updated 10 days ago by Gaurav Sanwal | 6 likes | 14 comments

Like page Watch More actions...

Added by Confluence Cloud

12 1 1 1 1 reply 1 month ago

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Friday, June 16th

Yogesh Bhagat 10:48 AM

@channel I just reviewed the progress all of you have made for I90. All of you are real stars, building futuristic product at unprecedented speed is not easy. More important is kind of features planned in future. It's dream come true for me . In last 20+ years, this is best technical product I have seen getting built by me , my peers or team around me . Special thanks to leaders like @Lalo, Elie @Akavaram, Rajesh who are spearheading adoption of it . Continue to spread knowledge and take this product to height where we can open source this to tech community outside CVS. Best thing - I heard today is Apollo was amazed to see features which we have built ourself and they were thinking of building it for commercial product . Thanks to Apigee team for their contribution . @Siva, Amaresh @Weber, Tracey D @Walsh, Patrick J @Beyers, James M @Mahajan, Mayur - Please share the clip , deck and detail on this channel . (edited)

14 1 1 2 1 1 1 1 1

2 replies Last reply 14 days ago

Appendix

Experience API Framework - Features

- ✓ No Code
- ✓ Federated – Super Graph
- ✓ Rest and GraphQL Support
- ✓ Dynamic Orchestration Based on Query – Framework
- ✓ Domain-driven design
- ✓ Telemetry, Observability, Traceability & Logging Framework
- ✓ Automate Member events (*)
- ✓ Automate VALET Metrics – Measure Success of Domain
- ✓ Client Development Portal (*)
- ✓ Zero time Deployment
- ✓ Test Automation (*)
- ✓ Schema Registry (*)
- ✓ Schema Check (*)
- ✓ Open Source Framework and building extension
- ✓ Scaffolding framework
- ✓ Create Swagger for rest API (*)
- ✓ ML for predictive cache based customer Journey (*)

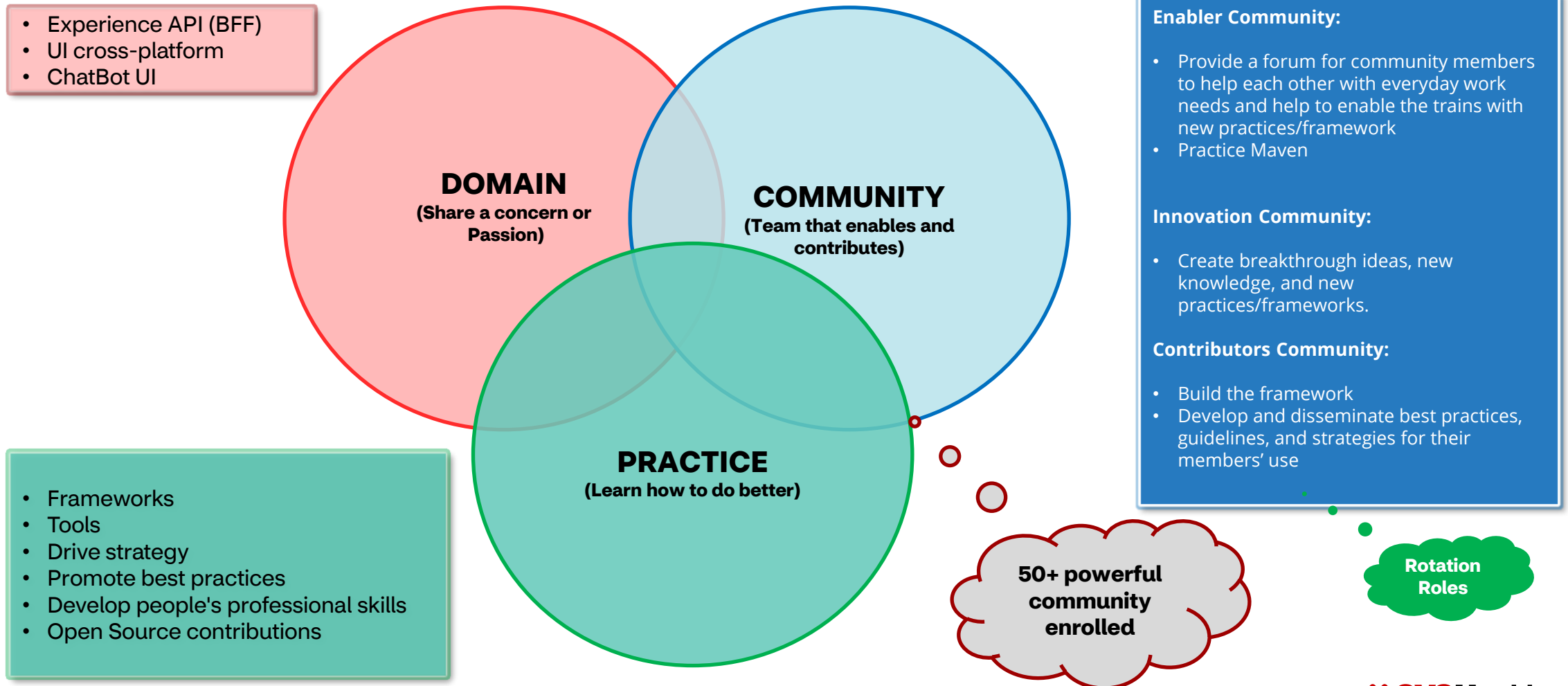
Aggregation Framework – Approach

While graphql efficiently solves the content curation problem, it comes with a cost. As the experience get complex , graphql is expected to plan and fetch responses in parallel or sequential across various capabilities/domain services and flatten the response. It is essential we return responses in sub-second and handle partial failures efficiently. There is a need to introduce guard rails around the federation framework , hence an efficient aggregation framework

- **Aggregation Framework is expected to**
 - Allow client development teams to register their queries, in other words profile efficient queries in a separate registry. This allows predictability and guaranteed response times.
 - End to End instrumentation
 - **Logging** - Catching failures and partial failures allows developers to further tune the application and handle experiences gracefully
 - **Metrics** - Although the framework is smart and solves technology problems , it is key we cater to Business and SRE Slo's. framework should ensure VALET metrics are provided Volume, Availability, Latency, Errors, Tickets
 - **Tracing** - Omni Pharmacy experiences are complex in nature and involves invoking services across multiple lines of business , capabilities and enterprise domains. It is key the framework allows tracing system end2end strictly adhere to open telemetry standards.
 - **Auditing** - Framework should consider using an auditing framework this ensures we provide value adds like Analytics, Forensics

Digital Community of Practice

Community of Practice (CoP) is a group of engineers & architects who share a concern, a set of problems, or a passion about a topic, and who deepen their knowledge and expertise by interacting on an ongoing basis



Experience API (BFF) – Problem Statement

- ✓ Inability to view customer journey and transactions at granular level using our current observability framework
- ✓ Teams spend significant time in building experiences over building capabilities and functional expertise in these areas
- ✓ Inconsistent monitoring and alerts dashboard, doesn't meet VALET metrics standards
- ✓ Delayed time to market , as teams spend developing granular api's
- ✓ Reusability of api's would mean clients over-fetching or under fetching resources
- ✓ Inability to detect source site of transaction to support white label and personalization
- ✓ Difficult to support monetization of API as Client registration and monitoring is after thought
- ✓ Missing API documentation and inconsistent rest standards across teams
- ✓ Inconsistent exception handling , disjointed experience while triaging issues for maintenance and system failures
- ✓ Different failures handling logic across teams, during scheduled maintenance, temporary external system failure or unexpected system outage.
- ✓ Absence of unified caching strategy , teams build cache context in silo's
- ✓ Business and data transformation logic seeps into the experience layer
- ✓ Token support is fragmented and supports only single line of business, hence significant number of api's catering to LOB's

Experience API (BFF) –

Problem Statement:

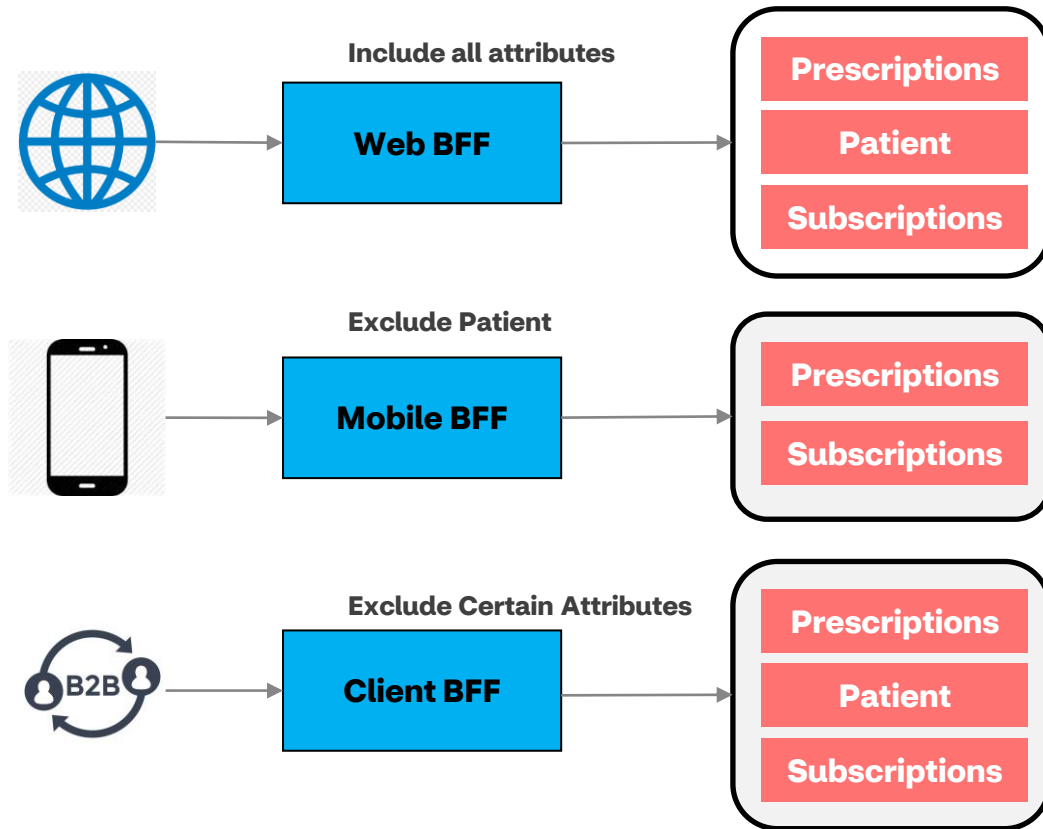
In pursuit of building thin Client applications, Pharmacy teams leverage Back-end-for-frontend aka BFF architecture. BFF helps curating backend server response in terms of

- Aggregation
- Sorting
- Filtering
- Cache-management
- Authorization
- CMS integration

Extensive research suggest, integrating graphql could potentially solve in terms of reducing the number of BFF's developed specific to a client experience

BFF Approach

Onboarding



Federation Approach

